ATIF ASSISTANT - MASTER PLAN

Local-First Human+AI Hybrid System

Dashing, Practical & Affordable

1. CORE PRINCIPLES

- Local-first: runs fully offline; no mandatory cloud
- Privacy-first: read-only by default, propose-then-approve
- Phase-independent: each phase works standalone
- Evidence-based: FACT/INFERENCE/ASSUMPTION/UNKNOWN/PREDICTION
- Advisory-only: speaker ID never auto-stored
- Human+AI hybrid: context-aware, emotionally aware

2. QUICK VISION

Auto-connect to Google Drive/Gmail/WhatsApp, read/update memory, auto-extract tasks from chats, learn per-person context (voice/tone/face), spatial mapping/home 3D, object/material awareness, capability estimation.

3. PHASES (EASY -> HARD)

P0: Baseline Hardening

Extend existing atif-assistant; CRUD solid; tests green (392/0)

P1: Connectors (Read-Only)

Drive/Gmail/WhatsApp/local files; propose-then-approve; dedup

P2: Task Extraction & Todos

Parse chats, detect intent ('come', 'create', 'do'), auto-add to todo

P3: Memory Enrichment

Facts/episodes/rules/learned; auto-update from data; prune

P4: Perception (Glasses)

Mic/cam, voice ID, tone, face recognition, on-device

P5: Spatial/Home Mapping

Location, objects, shapes/materials, home model

P6: Reasoning + Capability

Strength vs object, optimal way; human context

P7: Offline LLM Optimization

Small models, chunking, piecewise merge

P8: 3D/Home Export

Generate 3D from collected data

P9: Polish & Deploy Local

Self-hosted, no cloud deps

4. AFFORDABLE GLASSES SHORTLIST

Raspberry Pi Zero 2W + camera + mic | ~$35-50 | On-device, WiFi, flexible

ESP32-S3-EYE / ESP32-CAM + mic | ~$15-30 | Ultra-cheap, needs host

Waveshare RP2040/others | Varies | Low power

Existing smart glasses (basic) | ~$40-70 used | Quick start

5. DATA WITHOUT CLOUD

SQLite + FTS5, local embeddings (small models), tiering (recent/archive), compression, dedup, pruning. Keep hot data small; index by relevance.